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A Dataset Details

We initially collected a starting set of a thousand problems and natural language solutions by hiring freelance contractors on Upwork ([upwork.com](https://www.upwork.com)). We then worked with Surge AI (surgehq.ai), an NLP data labeling platform, to scale up our data collection. After collecting the full dataset, we asked workers to re-solve all problems, with no workers re-solving problems they originally wrote. We checked whether their final answers agreed with the original solutions, and any problems that produced disagreements were either repaired or discarded. We then performed another round of agreement checks on a smaller subset of problems, finding that 1.7% of problems still produce disagreements among contractors. We estimate this to be the fraction of problems that contain breaking errors or ambiguities. It is possible that a larger percentage of problems contain subtle errors.

To assist contractors with writing questions, we provided seed questions automatically generated from a few-shot prompted 175B GPT-3 model. Contractors were allowed to use those seed questions directly, to use them as inspiration and make modifications, or to come up with their own questions entirely. We instructed contractors to be as descriptive as possible in their solutions, and to not re-use problem settings or templates between different questions. To ensure contractors were not re-using problem templates, we computed pairwise similarity scores between problems and used this to provide feedback to contractors.